Modern Testing Reality

Faster Releases, Hardened Apps, and Less Time to Debug

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Agenda

01 Industry Trends

How AI, security, and scale are reshaping modern testing

02 Testing in a Secured World

Why hardened apps break traditional testing

03 Faster Decisions

Turning test results into faster, smarter decisions

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What's Happening?

QA teams are expected to do more with less.

AI is accelerating feature development within existing enterprise applications, increasing the pace and volume of changes that must be validated.

Coverage must scale with growing complexity.

Applications are expected to work flawlessly across a growing set of device, OS, browser, and environment combinations customers rely on daily.

Security protections break traditional testing.

When applications are protected against reverse engineering, automation testing platforms no longer have access, and testing gets blocked.

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Every Release Carries Business Risk

In an app-driven economy, every release carries revenue, customer trust, and compliance risk.

User Experience Testing

Functional

Reliability Drives Retention

- **88%** of users are less likely to return after a poor experience. [1]

Performant

The Cost of Slow Performance

- **~50%** of users abandon a page after a 2-second delay. [2]

Accessible

Accessibility Is the Rule, Not the Exception

- **88%** of analyzed websites not compliant with ADA standards. [3]

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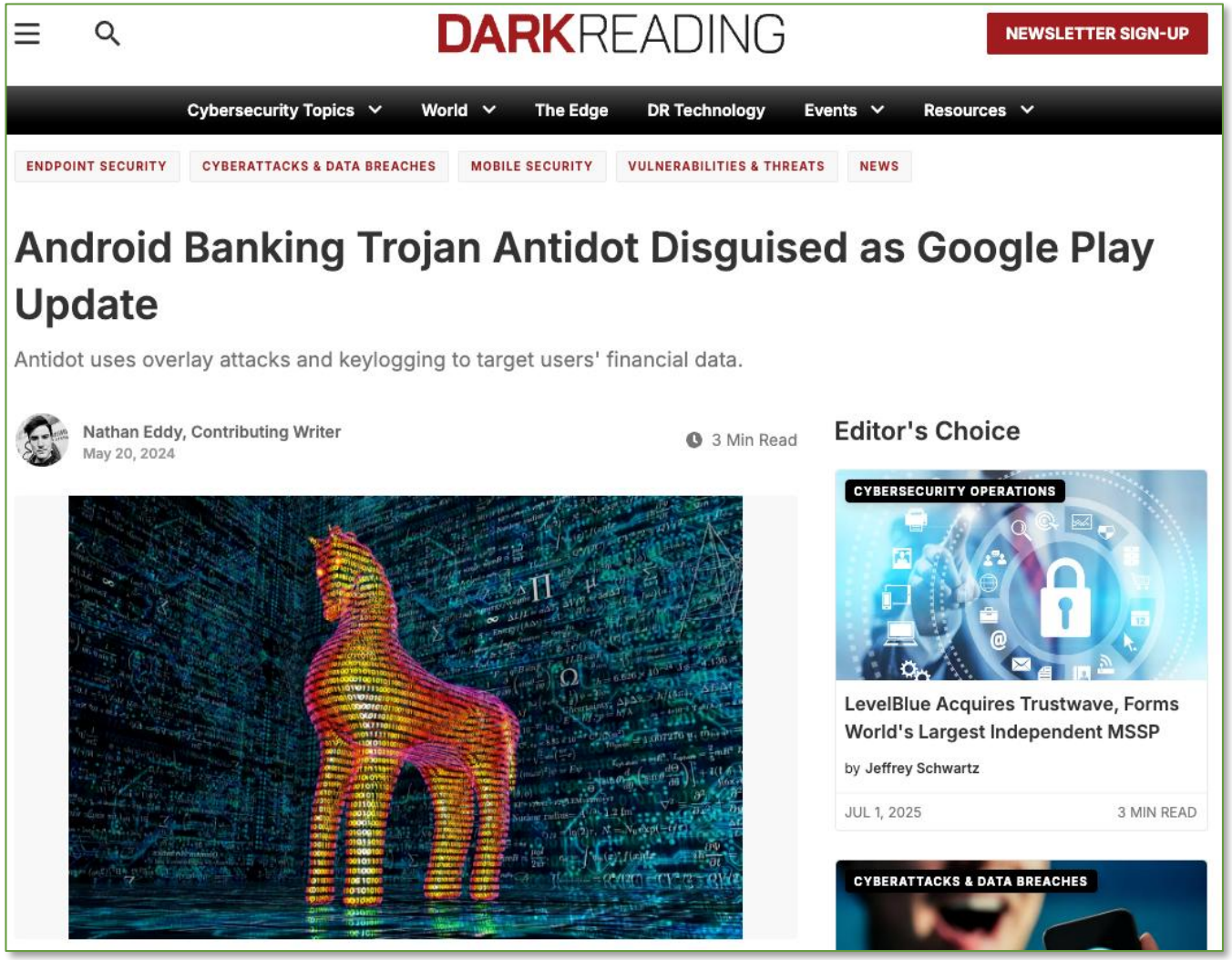
Fake Job Recruitment App Delivers Banking Malware

- Phishing campaign targeted banking apps via a fake job “CRM” app
- Malware injected overlays at runtime to steal user credentials
- No backend breach, attacks happened entirely on the client side



DARKREADING

NEWSLETTER SIGN-UP

Cybersecurity Topics | World | The Edge | DR Technology | Events | Resources

ENDPOINT SECURITY | CYBERATTACKS & DATA BREACHES | MOBILE SECURITY | VULNERABILITIES & THREATS | NEWS

Android Banking Trojan Antidot Disguised as Google Play Update

Antidot uses overlay attacks and keylogging to target users' financial data.

Nathan Eddy, Contributing Writer
May 20, 2024

3 Min Read

Editor's Choice

CYBERSECURITY OPERATIONS

LevelBlue Acquires Trustwave, Forms World's Largest Independent MSSP

by Jeffrey Schwartz

JUL 1, 2025 | 3 MIN READ

CYBERATTACKS & DATA BREACHES

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Parler iOS App Reverse Engineered

- iOS app reverse engineered to expose private APIs
- Undocumented endpoints used to scrape data and bypass auth
- Client-side logic enabled large-scale data extraction

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"But We Already Test…" — Is That Enough?

Testing isn’t the goal. Delivering flawless user experiences is.

Are you testing the right things?

Success isn’t just about how much of the app you test—it’s about focusing on the moments that matter most, where a single failure can cost you a customer.

Are you moving at the speed of your users?

Your customers upgrade fast. If your apps aren’t ready for the latest devices and OS versions on day one, you risk losing them to competitors who are.

Are you finding issues early enough?

When defects surface just before launch—or worse, in production—they become far more expensive to fix and significantly increase release risk.

The real question isn’t “*are you testing?*”—it’s whether your testing helps you release with confidence.

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Testing In a Secured World

Why hardened applications break traditional testing

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Mobile Applications Are Your Key To Success

$935B

Mobile apps are expected to generate over $935 billion in revenue in 2024.

5M

The Apple App Store has 1.96 million apps and Google Play Store has 2.87 million apps.

10

Average mobile users uses 10 apps per day and 30 apps each month.

1. Mobile App Download Statistics & Usage Statistics (2024)
2. Top 10 Benefits of Mobile Apps – Adobe\

Attacks by Industry

Healthcare – 78.5% **Automotive – 86%** **FinServ – 87.5%** **Telecom – 91%**

As apps become more critical to business and user outcomes,
they become the primary target for attackers.

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Why Traditional Security Testing Demands More

Let’s paint a scenario…

A financial services company had everything in place—
SAST on every commit, DAST before every release.

On paper, everything looked secure.

But once the app reached real users, the risks changed.

The attacks weren’t targeting code or APIs—
they targeted the running application itself.

Nothing was flagged—not because anything was missed,
but because this is a different problem.

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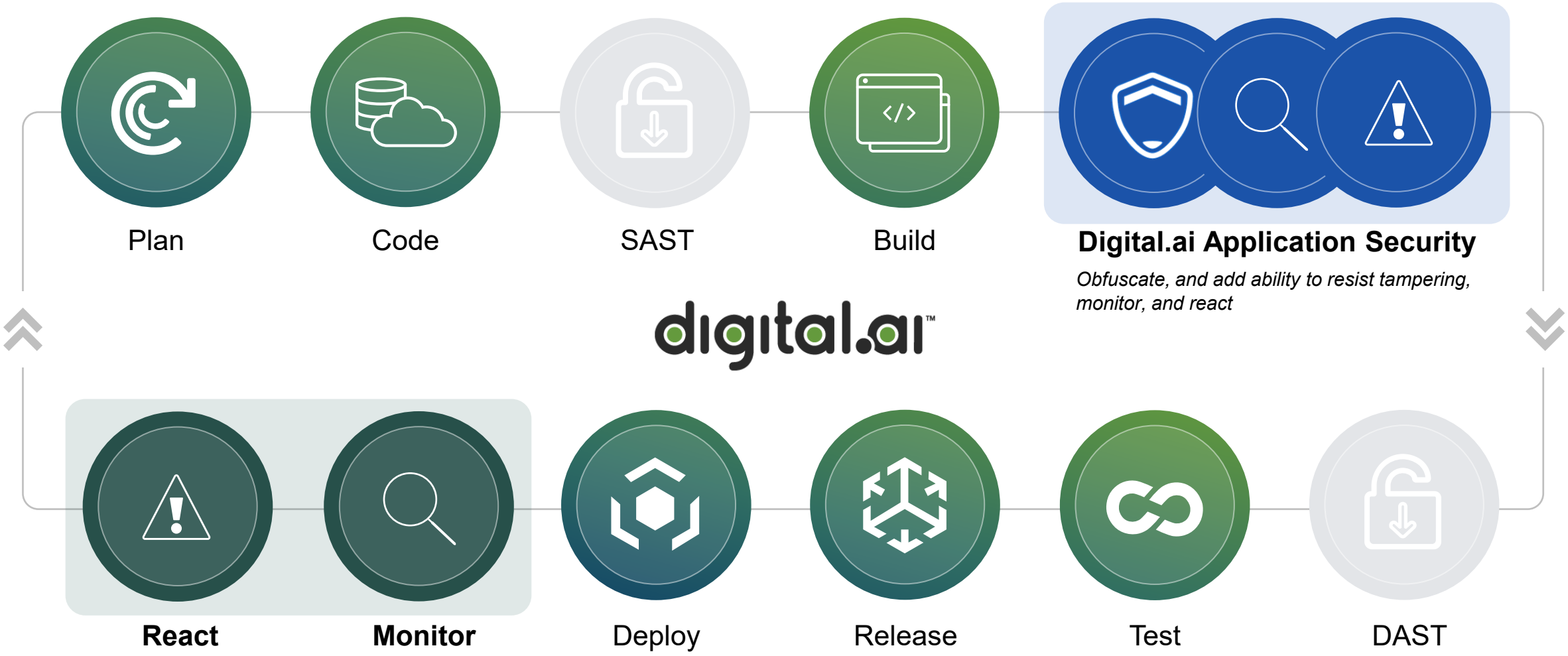
Where Security Sits in the SDLC



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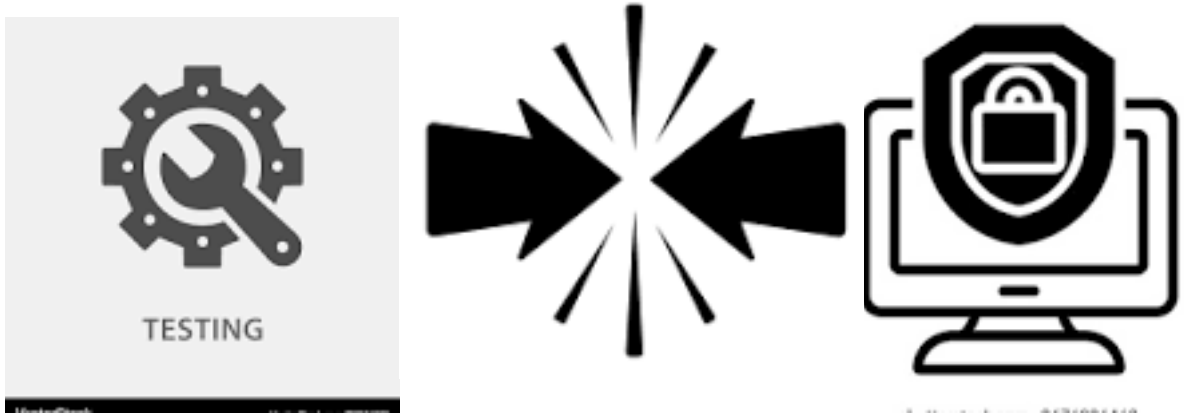
The Problem Nobody Talks About

- Enterprises protect mobile apps by adding hardening, anti-tamper, and runtime protection
- QA teams rely on automation to accelerate release cycles (CI/CD)
- The moment hardening kicks in, test platforms look like malicious interference



RESULT: your automated tests fail, do end-to-end testing manually, or you ship without full protection

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Why This Happens

- Hardened apps detect debuggers, emulators, injection tools, and simulated environments
- They behave exactly as designed – stop executing to prevent reverse engineering
- QA and Security are stuck in a loop:
  - QA wants automation to run
  - Security wants maximum protection
  - Both can’t be true without integration

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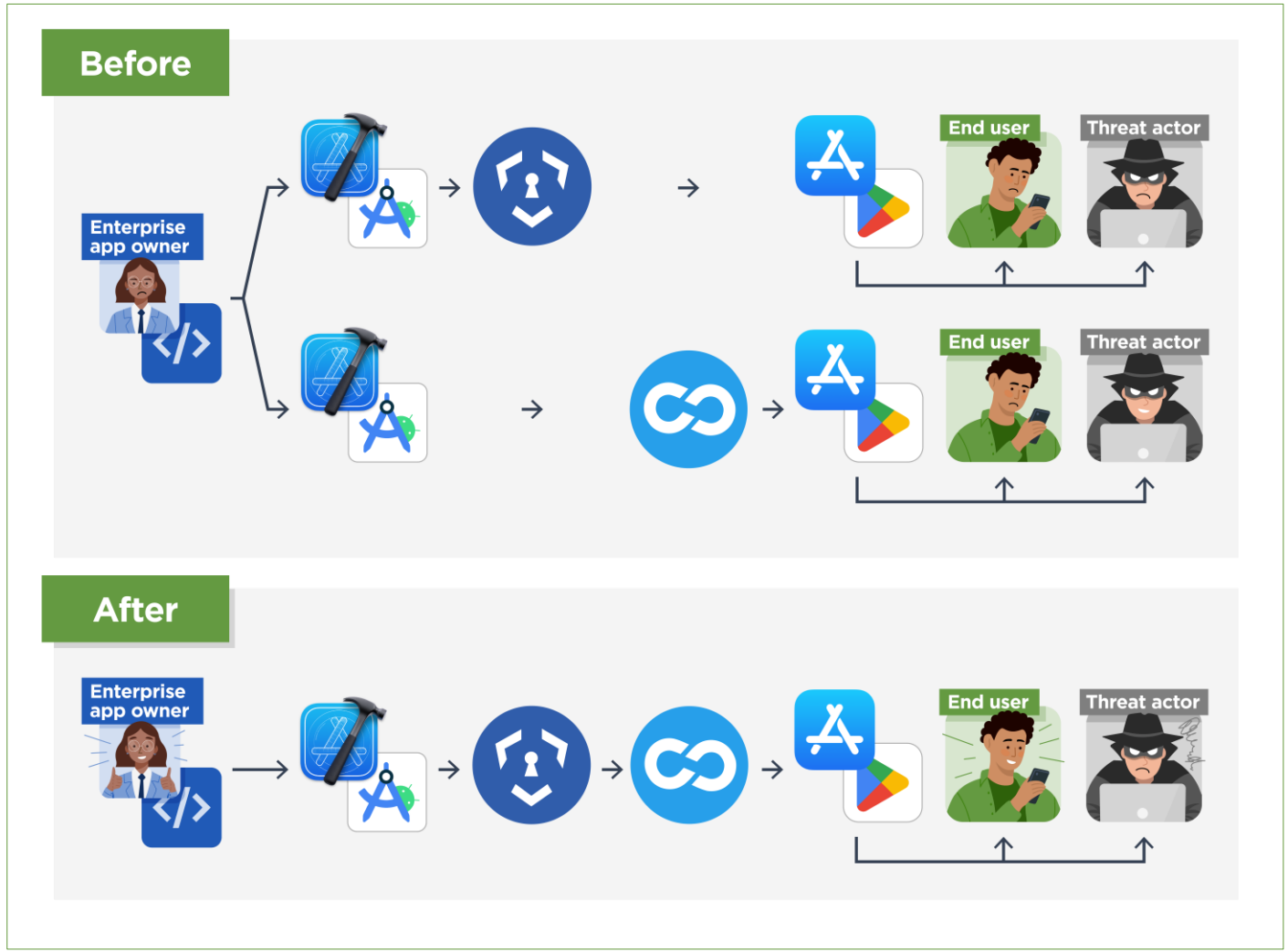
Your Tests Called – They're Tired Of Getting Blocked

- Our AppSec and Testing platforms share trust signals
- During testing, the hardened app recognizes Digital.ai Testing as a safe environment
- In production, protection return automatically to full enforcement



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Introducing: Digital.ai Testing

A centralized device farm for manual and automated testing across mobile devices and desktop browsers.

Test across devices, OS versions, and environments—giving leaders the visibility they need to make confident release decisions.

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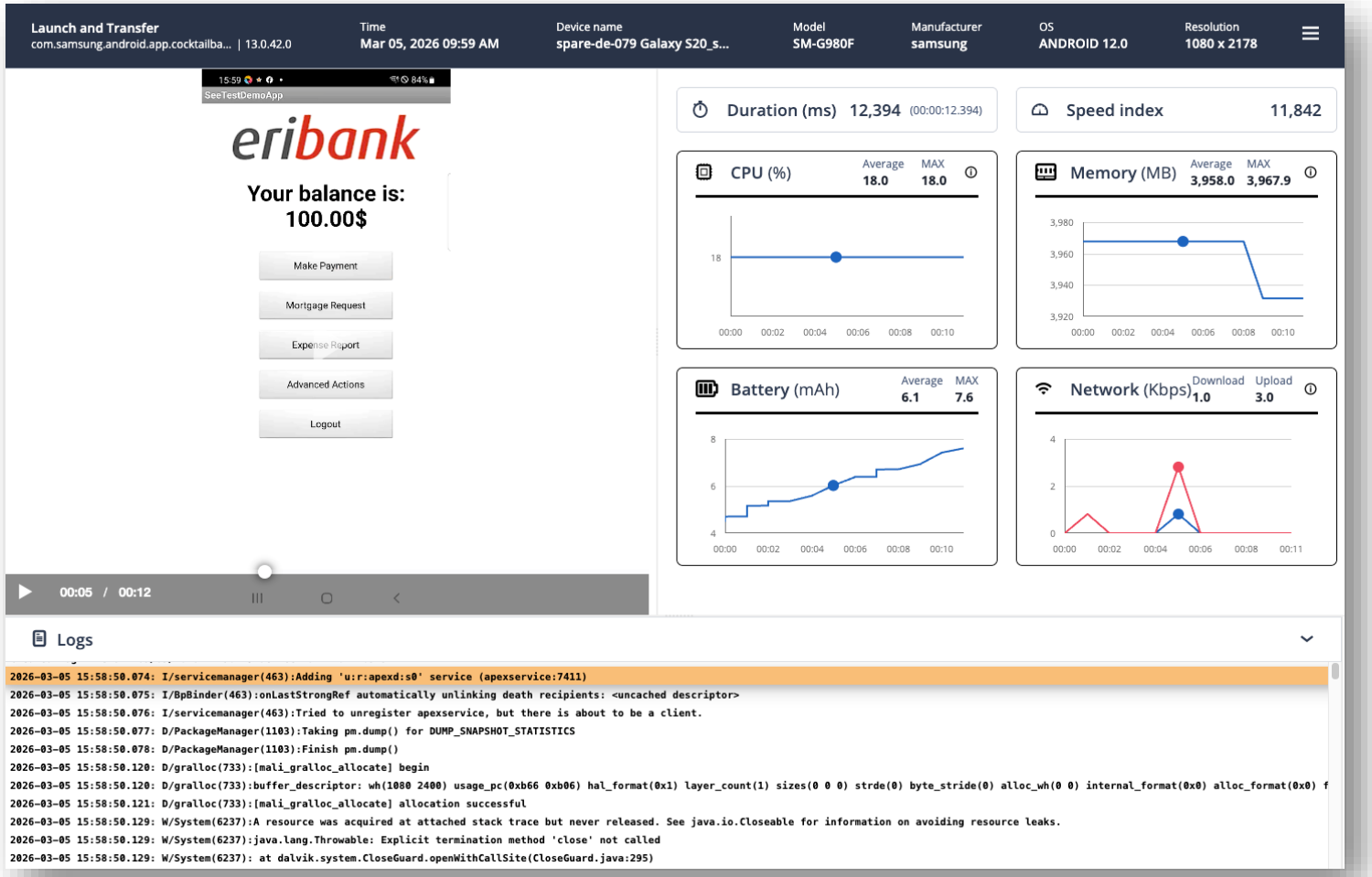
Protections Can Degrade Performance

- App protections can impact performance
- Hard to understand what changed—and why
- Measure real device metrics (CPU, memory, battery)
- Compare behavior across protection levels
- Secure apps without breaking the user experience



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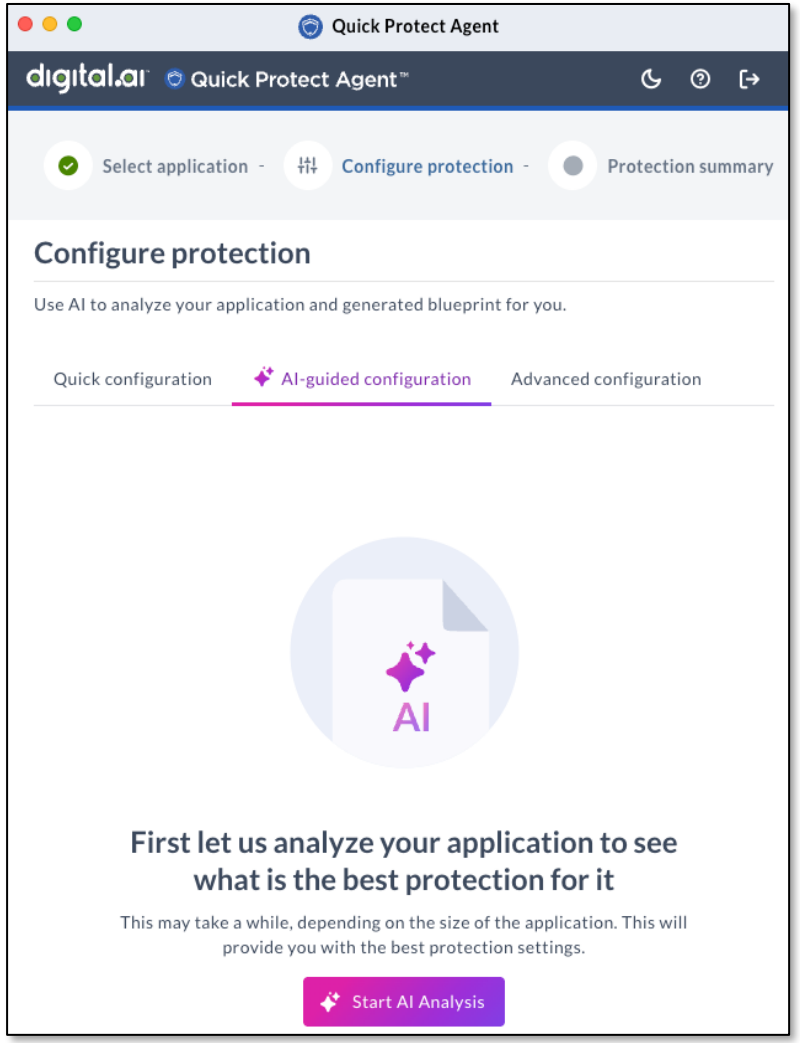
To Fight AI-fueled Attacks, You Need AI-fueled Protections

- Secure your app in minutes—not weeks
- AI-guided protection based on real application behavior
- No coding or scripting required to get started
- Flexible: start simple, customize as needed



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From Risk to Confidence

From Testing Bottlenecks to Secure, Faster Releases.

An identity and access management provider needed to **test hardened apps without slowing releases** or **compromising security**.

Challenge

Tight security controls meant teams could only test after hardening.

Testing was **manual and slow**, with little visibility into issues.

When builds failed, teams had to **restart from scratch**, delaying releases and raising costs.

Solution

Digital.ai Testing enabled **automated validation on hardened builds.**

Teams could verify **functionality**, **security**, and **performance faster**, without changing their process.

Outcome

- **80% faster** post-production validation
- **~25% lower** tooling effort and cost
- **Quicker, more confident releases** for secure applications protecting brand reputation

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Anonymized for Story Telling

Faster Decisions

RCA turns test results into actionable intelligence

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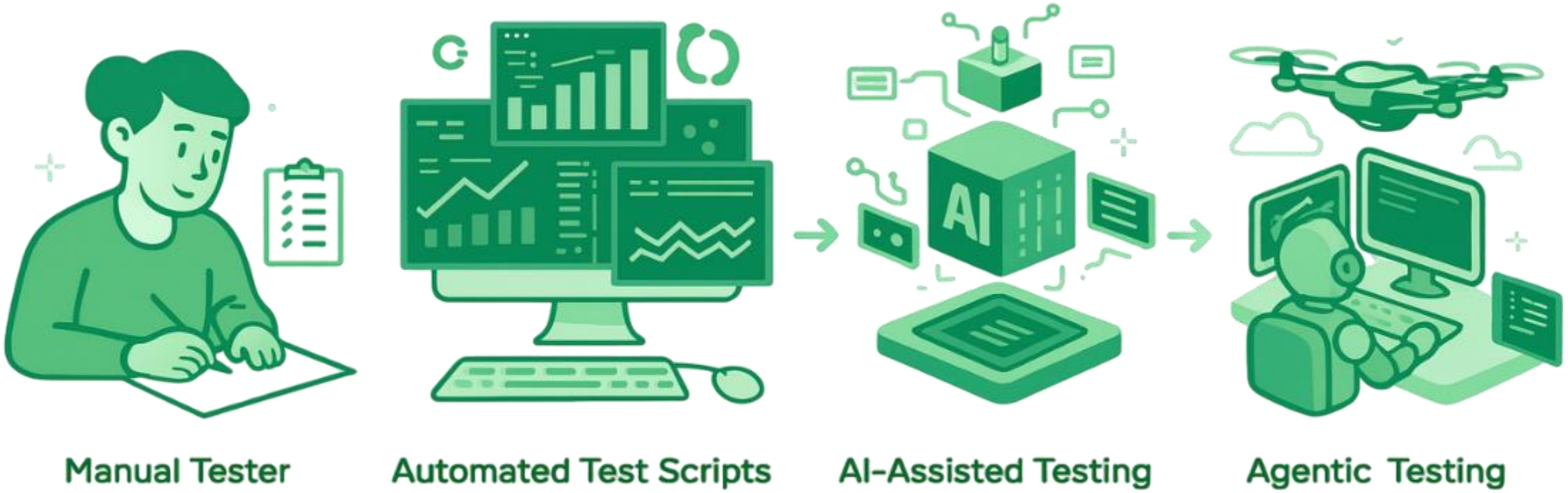
From Manual to Agentic: The New Era of Testing



How testing evolved and where the gap emerged

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Agentic AI for Testing Got Split in Two

Both matter, but over-indexing on one creates risk

Which one is right?

Generating More Tests

- Explosion of AI-generated test cases
- Focus on coverage and speed
- Easier test creation than ever before

More execution ≠ more confidence

Understanding Failures

- Failures increasing with scale
- Hard to separate:
  - real defects
  - flaky tests
  - environment issues

Insight becomes the bottleneck

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2025 at a Glance

What the data tells us*

7.1M+
Test Executions

1M+
Device Sessions

1.2M+
Browser Sessions

**Based on platform usage across SaaS*

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Why Root Cause Analysis Holds Teams Back

Every minute counts.

Testers spend on average **28 minutes** just identifying root cause of a single issue.

RCA is a bottleneck.

Test selection, prioritization, and RCA consistently rank among the **top three challenges.**

The cost is staggering.

In 2025, 2.5 million customer tests failed, costing **1.2 million hours** in investigations.

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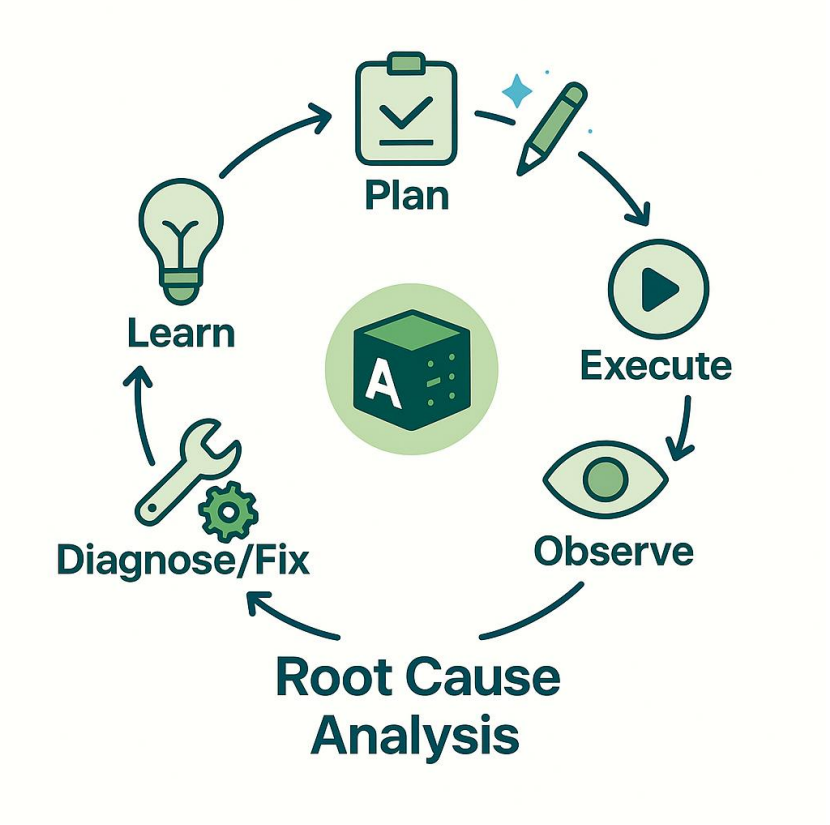
1. [Slack] Flaky Tests at Scale
2. [OpenText] The Future of Quality

Root Cause Analysis is THE Foundation

Root Cause Analysis is the **decision engine** of modern testing:

- Finds the **true cause**, not just symptoms
- Surfaces **patterns across failures**
- Guides the **next best action**



Without RCA, testing is blind.
With it, teams move faster—with confidence.

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The Shift from Reactive to Proactive Root Cause Analysis

1 Filter Test Results

2 Open Failed Test

3 Find Failed Step

4 Observe Video

5 Go through logs

6 Root Cause Found!

Tedious and ***labor-intensive*** review.

1 Tests are classified and grouped by root cause

2 Insights & recommendations for each error category

Instant root cause for all failures.

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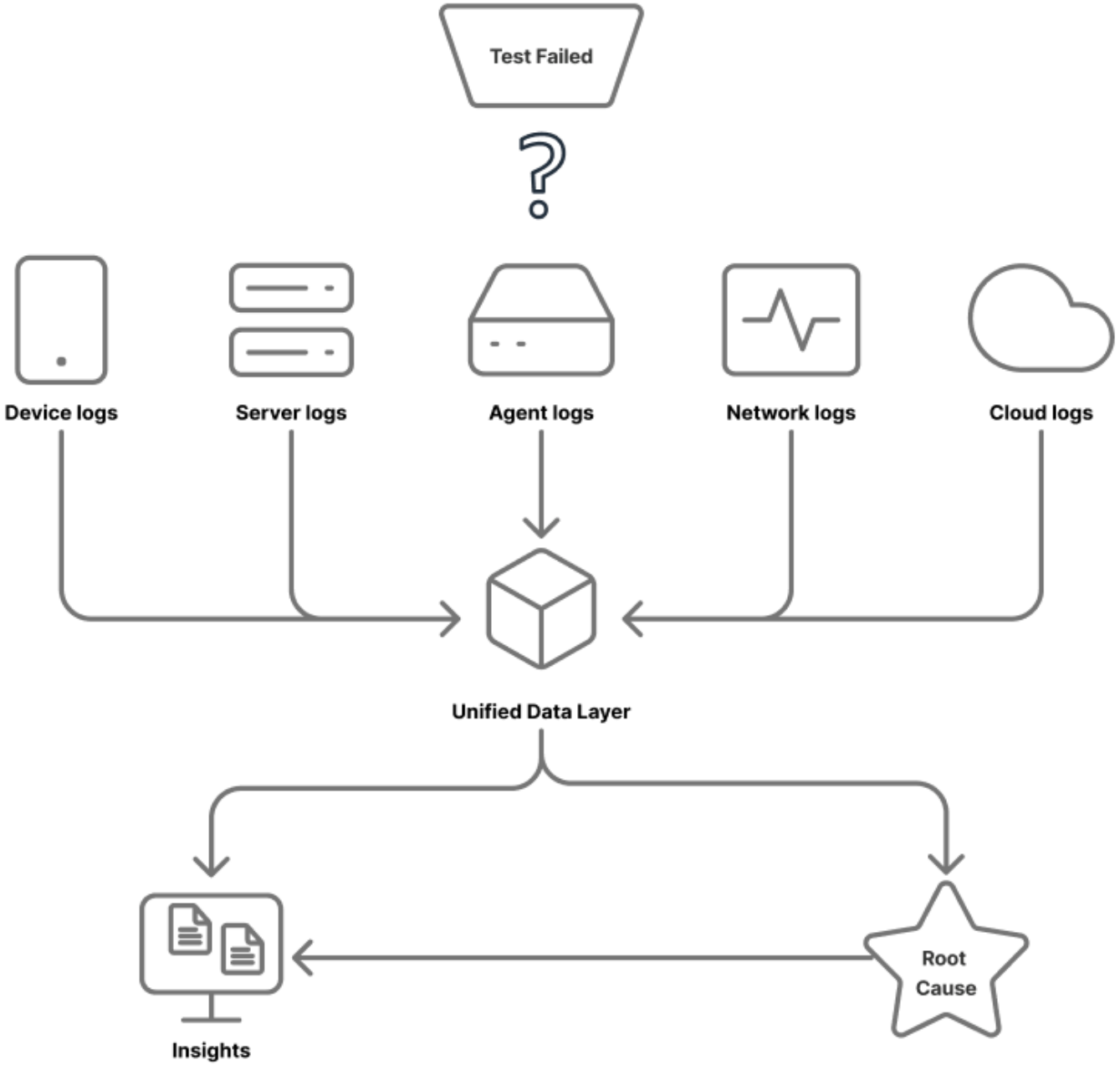
AI agents need adequate data to make good decisions



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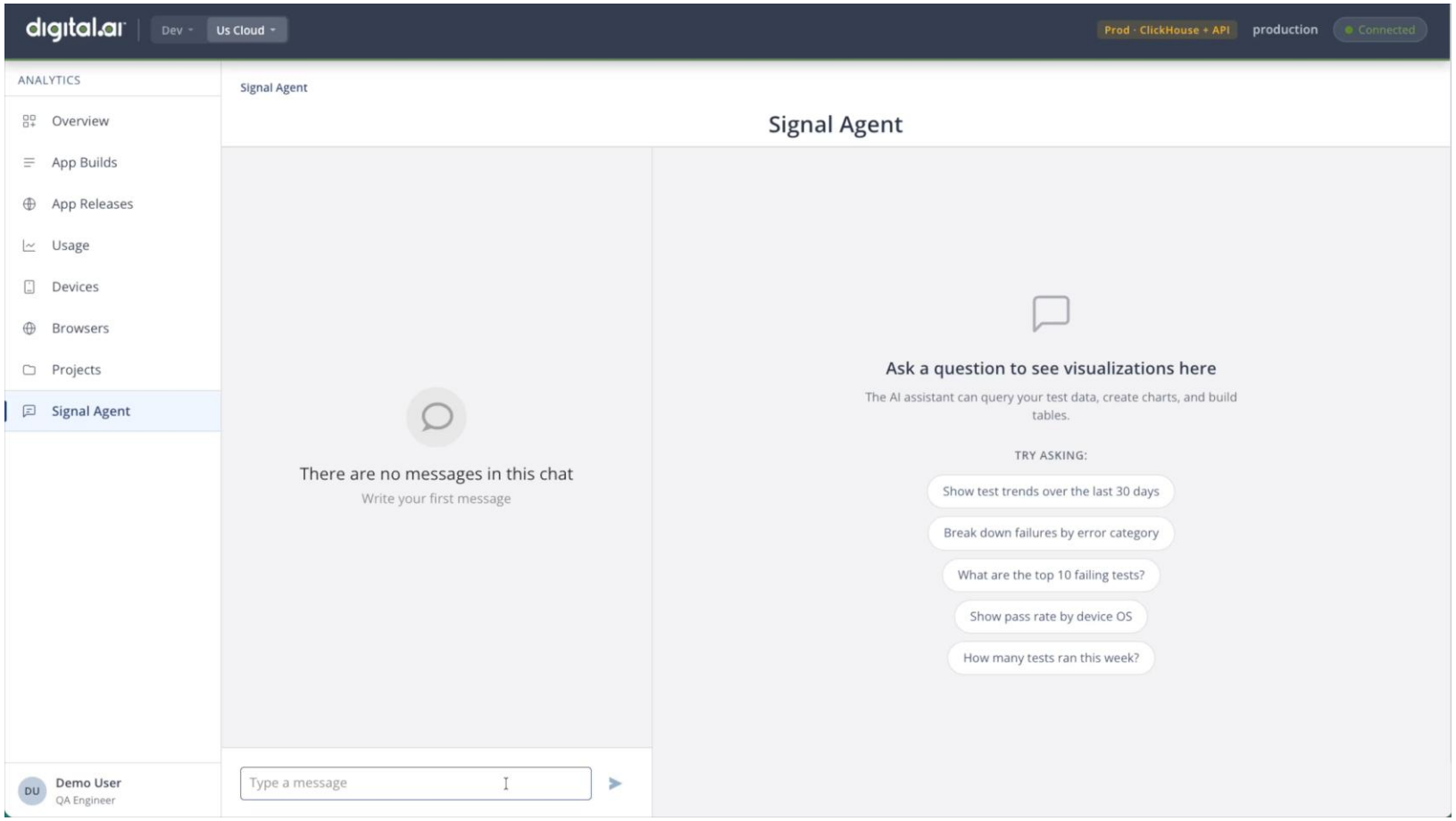
It Goes Beyond Test Execution Results

Test results tell you what happened.

Data tells you what to do about it.



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What Every Team Wants

Confidence

Make release decisions with clarity—knowing your app works as expected and is secure in the real-world

Prevention

Catch issues before release—preventing defects, and security risks from reaching users

Speed

Release high-quality applications faster without increasing risk

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The New Reality

Security is no longer separate from testing

Protected apps are the new baseline—not an edge case

The bottleneck has shifted from execution → understanding

Teams that can’t explain failures quickly will fall behind

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Thank You!

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